Physics-Informed Machine-Learning Inversion of Chandrayaan-3 RAMBHA-LP Langmuir Probe Sweeps

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The RAMBHA-LP Langmuir probe on the Chandrayaan-3 Vikram lander recorded current–voltage sweeps of the near-surface lunar plasma near the south pole between 24 August and 3 September 2023. Electron density, electron temperature and plasma potential are conventionally extracted by piecewise curve fitting, which fails on 44% of the 29,862 sweeps in the public rawB archive. We trained a physics-informed neural network on 40,000 synthetic sweeps from an orbital-motion-limited forward model, with noise and parameter ranges measured from the real archive, and a loss that also penalises failure to reconstruct the input curve. 30% of the sweeps are clipped by the instrument and neither method can read them, so they are excluded; the network recovers 5,115 of the 5,641 unclipped classical failures (91%), raising usable coverage from 50.7% to 67.9%. With no ground truth on the Moon, we validate against an independent classical fit (no meaningful density offset; 4.4% per sweep over 15,147 unclipped sweeps), against the classical value on the other half of the same sweep for 2,157 recovered sweeps (5.3%), and by self-consistency between the rising and falling halves of each commanded sweep (2.4%, r = 0.992, 9,895 pairs); the daily medians also follow the mission-long density rise from 246 to 500 cm^{-3}, as the classical fit does. Median values agree with the published ranges for this instrument, except for density on the first three days. Physics-informed training worsens accuracy on synthetic data, yet on real sweeps it makes the two halves of each sweep 30% more consistent and removes a 5% density offset against the classical fit, showing that simulation alone cannot judge sim-to-real choices. A two-population extension with a working negative control finds no evidence for a second electron population, in agreement with the instrument team, and shows that the standard averaged-sweep diagnostic for multiple populations is not valid evidence on its own.